SOMASHEKAR NAKKA

Autonomous Robotics Engineer | Kalman Filtering (EKF/ESKF) | VIO/LIO SLAM | Resilient GNSS
+91 7032215859 | shekar3reddy4@gmail.com | linkedin.com/in/somashekar39 | github.com/SomaShekar2678

PROFESSIONAL SUMMARY

Sensor fusion and navigation engineer with 2.5 years of experience in state estimation for autonomous platforms. Builds IMU, stereo-camera and GNSS fusion using Extended Kalman Filter (EKF) and error-state Kalman filter (ESKF) for real-time 6-DoF pose, and integrates VINS-Fusion and FAST-LIO2 on ROS2 Humble. Brings anti-jam GNSS and a visual localization pipeline cut from 22s to 1s, for resilient navigation when GNSS is degraded. B.Tech ECE, NIT Delhi.

KEY RESULTS

- Cut end-to-end visual localization latency from 22s to 1s (about 20x).
- Extended GNSS interference mitigation from single-band L1 to multi-band L1/L2/L5.
- Delivered remaining useful life (RUL) estimates with 80% confidence intervals for a diesel engine fleet.

EXPERIENCE

AI Engineer, TAVAS Industries · Bhopal, MP · Jun 2025 to Present

Robotics, Navigation & Sensor Fusion

- Developed multi-sensor fusion pipelines integrating IMU, stereo vision and GNSS using Extended Kalman Filter (EKF) and error-state Kalman filter (ESKF) state estimation for real-time 6-DoF pose.
- Integrated visual-inertial odometry (VINS-Fusion) and LiDAR-inertial odometry (FAST-LIO2) on ROS2 Humble for localization and mapping on autonomous platforms.
- Built multi-sensor synchronization and calibration workflows aligning LiDAR, stereo-camera and IMU streams for perception and state estimation.
- Built a visual localization pipeline as a GNSS-independent position source, cutting latency from 22s to 1s (about 20x).
- Developed GNSS anti-jamming and spoofing detection (L1/L2/L5) to keep navigation resilient under interference.

GNSS Signal Processing, Anti-Jamming & Resilient Navigation

- Designed the software architecture for a multi-antenna GNSS anti-jamming and interference mitigation system: AoA/DoA estimation, adaptive beamforming, spatial filtering, null steering and jammer localization on multi-channel IQ data.
- Extended GNSS signal processing from single-band L1 to multi-band L1/L2/L5 for interference detection and suppression across a broader spectrum.
- Implemented array signal processing algorithms: covariance estimation, MUSIC/ESPRIT direction of arrival (DoA) estimation, MVDR/LCMV beamforming, adaptive filtering (LMS/RLS) and spatial null steering for CRPA arrays.

Computer Vision & Real-Time Visual Localization

- Applied DINOv2, RoMa, RANSAC and FAISS for robust image correspondence and localization under viewpoint and appearance change.

Predictive Maintenance & Engine Health Monitoring (Diesel Engines)

- Architected an end-to-end prognostics and health management (PHM) pipeline fusing 1 Hz test-bench telemetry, stage sheets, oil analysis, in-service checks and teardown records into per-engine health, failure-mode and RUL outputs.

Software Engineer, Accenture · Hyderabad, TS · Oct 2022 to Nov 2023

- Trained and deployed Logistic Regression and Random Forest models to classify drug-adverse event pairs on structured healthcare data, using LIME for interpretability.
- Applied Generative AI and NLP to extract information from unstructured clinical-form sections such as medical histories and lab results.
- Built production FastAPI services for document extraction and report generation, with Celery and Redis for asynchronous OCR and text processing.
- Built document-processing pipelines for structured and semi-structured PDFs using PyPDF2, Tesseract and Pandas, with ETL into database records and statistical reports.

TECHNICAL SKILLS

- **Robotics & Sensor Fusion:** ROS2 Humble, IMU, LiDAR, Sensor Fusion, Kalman Filter (EKF, ESKF), State Estimation, 6-DoF Pose Estimation, SLAM, Visual-Inertial Odometry (VIO), LiDAR-Inertial Odometry (LIO), VINS-Fusion, FAST-LIO2, Multi-Sensor Synchronization & Calibration, Stereo Vision, Gazebo, RViz2
- **Signal Processing & GNSS:** Digital Signal Processing (DSP), GNSS Signal Processing, L1/L2/L5, Anti-Jamming, Interference Detection & Mitigation, Spoofing Detection, CRPA Arrays, Array Signal Processing, Adaptive Beamforming, Null Steering, AoA/Direction of Arrival (DoA) Estimation, MUSIC, ESPRIT, MVDR, LCMV, Adaptive Filtering (LMS, RLS), FFT, Spectral Analysis, Cyclostationary Analysis, PRN Correlation, IQ Processing
- **Computer Vision & Perception:** Computer Vision, OpenCV, DINOv2, XFeat, RoMa, Feature Extraction & Matching, Dense Matching, Image Retrieval, FAISS, RANSAC, Geometric Verification, Visual Localization
- **Programming & Tools:** Python, C/C++, SQL, Bash, JavaScript, PyTorch, TensorFlow, scikit-learn, statsmodels, SciPy, Pandas, FastAPI, Celery, Redis, PostgreSQL, Three.js, Git, Linux, Docker
- **FPGA, Embedded & Real-Time:** Xilinx (AMD) RFSoC, Kria KV260, Vivado, Vitis HLS, High-Level Synthesis (HLS), FPGA Acceleration, Hardware-Software Co-Design, Streaming Architectures, Fixed-Point Arithmetic, Embedded AI, Real-Time / Low-Latency Pipelines

- **Machine Learning & AI:** Computer Vision, Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Time-Series Forecasting, Reinforcement Learning, Explainable AI (LIME, permutation importance), Feature Engineering, Model Evaluation, Cross-Validation, Real-Time ML
- **Predictive Maintenance & Reliability:** Predictive Maintenance, Prognostics, Remaining Useful Life (RUL), Health Index / Health Score, Anomaly Detection, Isolation Forest, Change-Point Detection, Normal-Behaviour Modelling, Survival Analysis (Kaplan-Meier, Weibull, AFT), Mixed-Effects Models, Bayesian Updating, Gradient Boosting

EDUCATION & CERTIFICATIONS

- B.Tech, Electronics & Communication Engineering, National Institute of Technology (NIT) Delhi, 2018 to 2022
- PGP in Data Science & Machine Learning, MITx MicroMasters + Intellipaat
- Advanced Machine Learning Specialization, DeepLearning.AI (Coursera)